

**INDUSTRY INTERNSHIP
SUMMARY REPORT**

GenAI CRM Chatbot

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
Artificial Intelligence and Machine Learning

Submitted by

Asmita Sahani (23SCSE1180303)



**SCHOOL OF COMPUTING SCIENCE AND ENGINEERING
GREATER NOIDA, UTTAR PRADESH
SUMMER 2025-2026**

OFFER LETTER



Date: 10-07-2025

To,
Asmita Sahani

Subject: Internship Offer Letter

Dear Asmita,

We are pleased to offer you the opportunity to join Ravihans & Sons as GenAI Engineering Intern.

This letter confirms your internship, which is intended purely for training and learning purposes and should not be construed as an offer of full-time employment.

Internship Details:

- Role: GenAI Engineering Intern
- Start Date: 14-07-2025
- Duration: 1 month (extendable, depending on performance and project needs)
- Location: Remote
- Working Days: Monday to Friday, as per company guidelines
- Stipend: INR 20,000 (Base) + INR 5,000 (Variable, performance based) per month

On successful completion of your internship, you will be evaluated for a Pre-Placement Offer (PPO). The decision to extend such an offer will depend on your performance, feedback from your mentors, and the company's business requirements.

During your internship, you will be entrusted with access to company information, data, and project-related documents. All such material will remain the property of Ravihans & Sons. You are expected to maintain strict confidentiality and not share any company, client, or partner-related information with external parties without prior written approval.

Please confirm your acceptance by signing and returning a copy of this letter.

Warm Regards,


For Ravihans & Sons Private Limited

Radha Director
Authorized Signatory - Ravihans & Sons

Ravihans Aranyam, Ramnagar, Nainital, Uttarakhand 244715, India
<https://ravihanshospitality.in> | CIN: U79120DL2024PTC437266



Acknowledgement and Acceptance

I, Asmita Sahani, accept the terms and conditions outlined in this offer letter and agree to join Ravihans & Sons as an Intern.

Signature : Asmita

Date : 11-07-2025



Date: 15-08-2025

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Asmita Sahani**, Enrollment Number: 23131180139, a student of Bachelor of Technology in Computer Science Engineering (Artificial Intelligence & Machine Learning), Galgotias University, Uttar Pradesh has successfully completed her internship at Ravihans & Sons, New Delhi. Starting from 14th July to 14th August 2025. She was assigned tasks related to CRM Chatbot. The assigned tasks have been carried out with diligence with an acceptable level of perfection. We wish all the best for her future endeavours.


For Ravihans & Sons Private Limited

Director
Authorized Signatory - Ravihans &
Sons

Ravihans Aranyam, Ramnagar, Nainital, Uttarakhand 244715, India
<https://ravihanshospitality.in> | CIN: U79120DL2024PTC437266

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO
	Abstract	6
	List of Abbreviations	7
1	Introduction	8
	1.1 Objective of the project	
	1.2 Problem statement and research objectives	
	1.3 Description of Domain	
	1.4 A brief introduction about an organization.	
2	Technical Description	10
3	System Design	14
	3.1 General Architecture	
	3.2 Design Phase	
	3.2.1 Data flow diagram	
	3.2.2 UML Diagrams	
	3.3 Methodology	
4	System Implementation	19
5	Results and Discussions	21
6	Conclusion and Future Work	22
7	Appendices-	
	7.1 Source Code	
	7.2 Learning Experiences	
	7.3 SWOT Analysis	
		23

ABSTRACT

This section of the internship report provides a detailed explanation of the concepts, observations, and understanding gained during the training period. Throughout the internship, I was exposed to both theoretical ideas and practical workflows that helped me understand how real projects are executed in a professional environment. The learning experience combined technical knowledge with hands-on exposure, allowing me to connect classroom concepts with actual industry applications.

During the training, I spent time understanding the structure and functioning of the systems used in the organization. This included studying how different components interact with each other, how data flows within a system, and how customer facing technologies such as chatbots and CRM platforms work together to provide smooth services. Observing these workflows helped me understand the importance of well-designed architecture and the role of each module in the overall functioning of a project.

The internship also strengthened my analytical thinking. By going through various tasks, I learned how to break down problems, understand requirements, and analyze how different solutions may impact users and the organization. This process helped me develop a clearer understanding of system analysis and the importance of evaluating each step carefully before implementation.

In addition to technical concepts, I also gained exposure to industry practices such as documentation, planning, and reporting. I understood how documentation helps in maintaining clarity and consistency throughout the project lifecycle. Creating structured documents improved my ability to explain ideas clearly, organize information, and present findings in a professional manner.

Overall, this section highlights the knowledge I gained in terms of theoretical learning, workflow understanding, system analysis, and documentation practices. These learnings played a major role in enhancing my confidence and preparing me for future projects and professional responsibilities.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
NLU	Natural Language Understanding
NLG	Natural Language Generation
CRM	Customer Relationship Management
GenAI	Generative Artificial Intelligence
API	Application Programming Interface
DFD	Data Flow Diagram
UML	Unified Modeling Language
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation
DB	Database
FAQ	Frequently Asked Questions
SDK	Software Development Kit
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ML Model	Machine Learning Model
CSAT	Customer Satisfaction Score

CHAPTER 1

INTRODUCTION

Internships are one of the most important parts of a student's learning journey because they allow us to step outside the classroom and see how things actually work in a real company. During my internship at Ravihans & Sons Pvt. Ltd., I got the chance to understand how technology is being used in the hospitality industry and how different teams work together to provide smooth services to customers. Even though I was still learning, the internship helped me gain confidence and gave me a clearer idea of how professional environments function.

Before this internship, most of my knowledge came from textbooks, lectures, and online resources. But working with an actual organization showed me how theory is applied in day-to-day work. I was able to observe how people communicate, how tasks are assigned, how deadlines are handled, and how different tools are used to solve problems. This helped me understand that learning is not only about understanding concepts but also about how you use them in real situations.

Ravihans & Sons Pvt. Ltd. is part of the hospitality sector, and customer service is one of the most important parts of their work. During my internship, I realized how many questions and requests customers have every single day from bookings, availability, and check in details to activity timings, food preferences, and travel planning. Handling all these queries manually can be exhausting for staff, especially when the same questions keep coming repeatedly. Because of this, the company is exploring the use of CRM based chatbots and AI tools to make communication faster and more efficient.

One of the biggest things I learned during my training was how chatbots actually work behind the scenes. I always saw chatbots on websites but never understood what happened when someone typed a question. Through the training sessions and guidance, I learned how a chatbot identifies what the user is trying to ask, how it matches the question with the right response, and how it can even store or fetch information using a CRM system. Seeing this flow made me appreciate how many small steps go into making something that looks so simple from the outside.

I also learned about CRM systems and why they are important for businesses. CRM tools help companies keep track of customer information, bookings, preferences, and previous interactions. This helps them serve customers better and more personally. Observing how CRM works in the hospitality

sector made me understand how important it is for a company to stay organized, especially when dealing with many guests every day.

Apart from the technical aspects, the internship also taught me a lot about workplace discipline. Simple things like organizing files, maintaining notes, documenting work, and communicating clearly make a big difference. I understood why documentation is important it keeps everyone on the same page and helps new members understand what has already been done. Writing clear documentation also helped me improve how I express ideas and explain things step by step.

Another thing I appreciated about this internship was the chance to observe real workflows. I got to see how messages are handled, how customer data flows through different systems, and how staff members use different tools to solve customer problems. Understanding these workflows helped me develop a better sense of how systems are connected and how important it is to design things in a way that is easy for both staff and customers to use.

Even though the internship focused on CRM chatbots and related tools, it also gave me a general exposure to how a modern hospitality business uses technology. Whether it was understanding booking systems, digital communication methods, or automation tools, everything added to my understanding of how industries operate today. It helped me see that technology is not just about coding or machines it plays a major role in improving customer experience and making work more manageable for employees.

Overall, this internship helped me grow not only technically but also personally. I gained a much clearer idea of how organizations work, how teams communicate, and how technology fits into the bigger picture. The training period gave me valuable observations, helped me build confidence, and made me more prepared for future responsibilities. This introduction summarizes the main learning areas of my internship, including theoretical understanding, practical exposure, workflow observation, documentation experience, and the role of technology in real business environments.

CHAPTER 2

TECHNICAL DESCRIPTION

The technical part of my internship mainly involved understanding how CRM chatbots work, how they are designed, and how they can be connected with customer management systems used in the hospitality industry. Even though the chatbot development process may seem simple from the outside, it involves several concepts, tools, and workflows that need to work together smoothly. This section explains the technical understanding I gained during the training period, including the tools used, the structure of the chatbot, and the role of CRM and AI in improving customer interactions.

1. Chatbot Fundamentals

A chatbot is a computer based program that can communicate with users through text or voice. The main goal of a chatbot is to understand what the user wants and provide the correct response. During the internship, I learned that most modern chatbots are built using AI and Natural Language Processing (NLP), which help them understand user queries written in normal language.

The chatbot used in my training was built with Google Dialogflow. Dialogflow uses “intents,” which represent user questions or requests. Each intent contains training phrases that teach the chatbot how people might ask the same question in different ways. It also has predefined responses that are sent back to the user when the intent is matched. Understanding how intents work made it easier for me to see how chatbots are structured internally.

I also learned about entities, which are keywords or important pieces of information inside a user’s message. For example, if someone asks “What are the room prices for July?”, then “July” is an entity. Dialogflow extracts this entity and helps the system respond correctly. This made me understand that chatbots rely heavily on data extraction, classification, and context management.

2. Natural Language Processing (NLP)

NLP is the technology that helps chatbots understand human language. Even though I did not work directly on building NLP models, I learned the basic idea behind them. NLP helps the chatbot do the following:

- Identify the user’s intention

- Break down sentences
- Recognize important keywords
- Handle repeated or similar questions
- Provide appropriate responses based on context

Dialogflow uses built-in NLP, so I didn't need to write algorithms manually. However, observing the process helped me understand how AI tools simplify tasks that would otherwise require deep technical knowledge. For example, Dialogflow automatically understands that "book a room," "I need a room," and "Do you have rooms?" all point to the same meaning. This feature makes chatbots useful for customer service because people phrase their questions differently.

3. CRM (Customer Relationship Management) Basics

CRM systems are essential for businesses that handle customers regularly, especially in hospitality. During the internship, I was introduced to Zoho CRM, which the company uses to store and manage customer information. CRM helps companies keep track of:

- Customer details (name, phone number, email)
- Booking history
- Preferences and requirements
- Past queries
- Follow-up tasks

I learned how CRM and chatbots work together. When a customer asks a question, the chatbot can send or retrieve information from the CRM. For example:

- If a guest asks about their booking status, the chatbot can check the CRM.
- If someone asks for adventure activity timings, the chatbot can fetch the details from stored data.
- If a new customer interacts, the chatbot can save their query in the CRM for follow-up.

Understanding this flow helped me recognize why CRM integration is important in industries that receive many customer inquiries.

4. Tools and Technologies Used

a) Google Dialogflow

Dialogflow is used to create the chatbot. During the internship, I worked with:

- Intents

- Entities
- Training phrases
- Responses
- Contexts
- Integration options

Its simple user interface made it easy to practice without requiring advanced programming skills.

b) Zoho CRM

Zoho CRM helps store structured customer information. I learned how fields, modules, and records work. Even though I did not build a live connection between the chatbot and CRM, I learned the theory behind CRM integration.

c) Workflow & System Tools

I also learned basic diagramming tools like Draw.io to create:

- Data Flow Diagrams (DFD)
- System architecture
- Module-level diagrams
- Basic sequence diagrams

These diagrams helped me visualize how different components communicate internally.

5. High-Level Architecture Understanding

During the internship, I studied a simple architecture of how CRM chatbots function. The general workflow includes:

1. The user types a question.
2. The chatbot receives the message and processes it using NLP.
3. Dialogflow matches the message with the correct intent.
4. The system checks if CRM data is needed.
5. If needed, data is fetched from the CRM.
6. The chatbot prepares the answer.
7. The response is sent back to the user.

Learning this architecture gave me a clear picture of the data movement behind a single user message.

6. Practical Exposure During Training

Even though my role focused more on observing and learning rather than building full systems, I still got hands-on exposure to:

- Creating basic intents in Dialogflow
- Designing sample conversation flows
- Understanding fallback responses
- Observing CRM fields and modules
- Understanding how user queries are categorized
- Documenting technical processes

This gave me a practical foundation in chatbot workflows without needing to implement complex code.

7. Importance of This Technical Knowledge

The technical learning I gained during this internship will be useful in many future projects, not just in hospitality. Chatbots, AI tools, and CRM systems are becoming increasingly common in all industries. Understanding how they work will help me build stronger projects, communicate better with teams, and take on more technical challenges in the future.

CHAPTER 3

SYSTEM DESIGN

System design is one of the most important parts of any technology project because it explains how different components work together to achieve the overall goal. During my internship at Ravihans & Sons Pvt. Ltd., I learned how CRM chatbots are structured, how data flows through different modules, and how each part contributes to the complete system. This section describes the architecture, workflows, diagrams, and technical reasoning behind the CRM chatbot system that I studied during my training.

1. Understanding the Need for System Design

Before creating any technical solution, it is important to have a proper plan that explains how the system should behave. System design ensures that all components such as the chatbot interface, NLP engine, backend logic, and CRM work in a coordinated way. It also helps identify potential issues such as slow responses, unclear data handling, and missing information. During my internship, I learned that even small errors in design can lead to big problems later, especially in systems meant for customer interaction.

For example, if the chatbot does not clearly understand the user's intent, it may fetch the wrong information or fail to respond entirely. Likewise, if CRM fields are not mapped correctly, user queries may not get recorded, which affects customer follow-ups. Because of this, system design becomes the foundation of building a reliable and user-friendly CRM chatbot.

2. High-Level Architecture of the CRM Chatbot System

The simplest way to understand the chatbot is by looking at the high-level architecture. During the internship, I studied a general architecture that most service-based companies use:

1. User Interface Layer – where customers interact (website, chat widget, mobile app)
2. NLP & Intent Classification Layer (Dialogflow) – identifies what the user wants
3. Backend Processing Layer – handles logic, responses, and rules
4. CRM Layer (Zoho CRM) – stores customer data and retrieves booking information
5. Response Generation Layer – prepares the final reply

6. Delivery Layer – sends the response back to the user

This flow helped me understand how a simple message from a user goes through multiple steps before a final answer is delivered.

Example workflow:

User Question → Dialogflow Intent → Logic Processing → CRM Data → Chatbot Response

Even though this happens in seconds, the design behind it needs careful planning.

3. Data Flow Design

During the internship, I created and studied Data Flow Diagrams (DFD) to understand how data moves within the system.

DFD Level 0 (Context Diagram)

This diagram shows the chatbot as a single unit interacting with external systems:

- User sends input
- Chatbot System processes it
- CRM Database is consulted if needed
- Response is returned
- Logs are stored for analysis

It shows only the highest-level flow without internal details.

DFD Level 1 (Detailed Flow)

This level breaks the system into components:

- Input Processing Module reads the message
- Intent Detection Module identifies the meaning
- Entity Extraction Module finds keywords
- CRM Interface Module fetches or stores data
- Response Module prepares the output
- Feedback Module stores logs for improvement

Understanding DFDs helped me see how each part contributes to the overall functioning of the chatbot.

4. UML Diagrams Studied

UML diagrams helped me visualize how different parts of the system behave. I studied the following diagrams:

a. Use Case Diagram

Shows how the user interacts with the chatbot. The main use cases include:

- Ask for room availability
- Ask about adventure activities
- Request booking details
- Get travel package information
- Submit a query for follow-up

The actors are User, Chatbot, and CRM System.

b. Sequence Diagram

This diagram helped me understand the order in which different modules interact.

Example sequence for “Check Room Availability”:

1. User sends the message
2. Chatbot forwards the text to Dialogflow
3. Dialogflow identifies intent
4. Backend checks CRM or predefined database
5. Response is generated
6. Final answer is displayed to the user

c. Activity Diagram

Shows the workflow from start to end:

- User sends query
- System classifies it
- System checks knowledge base
- If data required → CRM API call
- Response generated
- If user asks again → loop back
- Interaction ends

These diagrams made it easier to visualize how the system handles multiple types of requests.

5. Internal Module-Level Design

Although I did not build full modules myself, I learned how they are structured from a design perspective:

a. Intent Classification Module

Uses NLP to identify user needs.

It includes:

- Training data
- Intent groups
- Mapping rules
- Fallback handling

b. Entity Extraction Module

Extracts keywords such as:

- Dates
- Room type
- Number of guests
- Location
- Activity names

This is important so the chatbot understands the details inside a message.

c. CRM Communication Module

Handles interactions with Zoho CRM.

Its responsibilities include:

- Reading CRM data
- Creating customer records
- Updating follow-ups
- Fetching booking information

Even though I did not code the CRM integration, studying its design helped me understand how chatbots use stored data.

d. Response Generation Module

Responsible for selecting the correct reply based on:

- Intent
- Entities
- CRM results
- Predefined message templates

Good design ensures replies sound natural and relevant.

6. Design Considerations Learned During the Internship

I learned that a good system design must consider factors such as:

a. Accuracy

The chatbot should correctly identify user questions.

Misclassification leads to wrong responses.

b. Speed

The system must respond quickly, especially in hospitality where customers expect instant answers.

c. Scalability

The design should support more intents as the company adds more services.

d. Security

CRM data contains customer information, so the system must handle it safely.

e. Error Handling

Fallback messages and re-prompting are necessary when the chatbot doesn't understand something.

f. User Experience

A friendly, clear, and conversational tone improves the user's interaction.

These considerations showed me how design decisions affect the final product.

7. Importance of System Design in CRM Chatbots

Through this internship, I understood that system design is not just a formal step but the backbone of the entire chatbot. Good design:

- prevents confusion
- ensures consistent results
- makes integration smoother
- reduces the workload of customer service staff
- creates a better experience for users
- supports future development

Understanding this helped me appreciate the technology behind tools that many people use every day without realizing how much planning goes into them.

CHAPTER 4

SYSTEM IMPLEMENTATION

The system implementation stage focuses on putting the planned chatbot design into a working structure. Although my internship involved more learning and observation than full-scale development, I did get hands-on experience with the basic steps used in building a CRM-based chatbot. This section explains the simplified implementation process I followed during my training.

1. Setting Up the Dialogflow Agent

The first step in the implementation process was creating an agent in Google Dialogflow, which serves as the central container for all chatbot settings. I learned how to set up the agent, select the default language, adjust basic configurations, and understand the interface. This step helped me become familiar with the platform and prepared me for working on intents and responses.

2. Creating Basic Intents

Intents are the foundation of the chatbot. They represent different types of questions a user may ask. During the internship, I created sample intents related to hospitality services, such as:

- Room availability
- Check-in and check-out details
- Activity timings
- Restaurant information

For each intent, I added training phrases so the chatbot could recognize different ways of asking the same question. I also added simple responses that the chatbot would send back. This helped me understand how intent detection influences the final user experience.

3. Adding Entities for Key Information

Entities help the chatbot extract important information from user messages. I created basic entities for dates, room types, and activity names. Dialogflow also provides built-in system entities, which made it

easier to work with common inputs like dates or numbers. Working with entities showed me how chatbots handle user-provided details.

4. Designing a Simple Conversation Flow

After building intents and entities, I created simple conversation flows. These flows defined how the chatbot should continue the conversation after responding. For example, if a user asked about room availability, the chatbot could follow up by asking for preferred dates. Designing these flows helped me understand how logical steps are used to guide users during a chat.

5. Observing CRM Integration Workflow

Although I did not code the actual integration, I studied how chatbots connect with CRM tools like Zoho CRM. I learned that:

- The chatbot identifies the user's query
- It sends a request to the CRM if data is needed
- CRM returns relevant information
- The chatbot formats the response and sends it back

This helped me understand how backend systems and chatbots work together to automate customer service tasks.

6. Testing and Improving Responses

Testing the chatbot in the Dialogflow simulator was an important part of the implementation. I tried different user questions to see if the chatbot responded correctly. Based on the results, I made changes to training phrases and responses. This taught me that implementation is an iterative process, where continuous refinement improves chatbot accuracy.

CHAPTER 5

RESULTS AND DISCUSSIONS

During my internship, I was able to understand how CRM-based chatbots function and how they improve customer service in the hospitality sector. Even though I worked on a simplified version of the system, the training and testing activities gave me clear insights into the performance and behavior of the chatbot.

One of the main results of the work was creating a functional set of basic intents in Dialogflow. These intents allowed the chatbot to identify common queries related to room availability, check-in times, activity details, and restaurant information. During testing, the chatbot responded accurately to most of the training phrases, which showed that the intent structure was set up correctly.

Another key result was noticing how intent detection improved after refining the training phrases. In the beginning, the chatbot sometimes confused similar questions, but after updating the phrases and adding more examples, the accuracy increased. This helped me understand the importance of good training data in chatbot performance.

I also gained a clear understanding of how CRM integration works. Although I did not implement the full integration myself, I learned how chatbots use CRM systems to fetch booking details, store user queries, and organize customer information. This helped me see how automated systems reduce manual workload for staff and make customer service faster.

From a user experience point of view, I observed how important clear and simple responses are. Well-designed fallback messages were especially useful when the chatbot failed to understand something. This helped maintain a smooth flow in the conversation and made the chatbot feel more helpful.

Overall, the results of the internship showed that even a basic chatbot can automate repeated questions and improve customer interaction. The discussions and observations helped me understand both the strengths and limitations of AI-based systems. This practical exposure gave me confidence and a clearer idea of how such systems are implemented in real businesses.

CHAPTER 6

CONCLUSION AND FUTURE WORK

While the basic structure of the CRM chatbot was explored during the internship, there is still a lot of scope for future development. Some areas that can be improved or expanded include:

- **Full CRM Integration:** The chatbot can be connected directly to Zoho CRM through APIs so it can fetch and update real booking data instead of using sample responses.
- **Advanced NLP Improvements:** Adding more training phrases, refining intents, and improving entity extraction will make the chatbot more accurate and natural.
- **Voice-Based Interaction:** A voice-enabled version of the chatbot can make customer support more accessible for users who prefer speaking instead of typing.
- **Automated Travel Recommendations:** AI features can be added to suggest trip plans, activities, and room options based on customer preferences.
- **Analytics Dashboard:** A reporting system can be built to track user queries, chatbot accuracy, and customer feedback for future improvements.

These enhancements can help the company build a smarter, more personalized, and more efficient chatbot system. With continued development, the chatbot can become an important tool for improving customer experience in the hospitality **sector**.

CHAPTER 7

APPENDICES

Source Code

Below is a sample portion of the chatbot intents and backend structure I worked on during the internship. This represents the basic logic used to match user queries and return appropriate responses. It also shows how entities and fallback messages were structured.

Sample Dialogflow Intent (Room Availability)

```
{  
  
  "intent": "CheckRoomAvailability",  
  
  "trainingPhrases": [  
  
    "Are rooms available?",  
  
    "Do you have any rooms right now?",  
  
    "Availability of rooms",  
  
    "Is any room free for today?"  
  
  ],  
  
  "responses": [  
  
    "Our rooms are currently available. May I know your preferred dates?",  
  
    "Yes, we have multiple options available. Please share your check-in and check-out dates."  
  
  ],  
  
  "parameters": {
```

```
"check_in": "sys.date",  
  
"check_out": "sys.date"  
  
}  
  
}
```

Sample Intent (Activity Timings)

```
{  
  
  "intent": "ActivityTimings",  
  
  "trainingPhrases": [  
  
    "What activities do you have?",  
  
    "Tell me the timings for adventure activities",  
  
    "When does the safari start?"  
  
  ],  
  
  "responses": [  
  
    "Adventure activities start from 9 AM to 5 PM.",  
  
    "The jungle safari begins at 6 AM. Please contact reception for bookings."  
  
  ]  
  
}
```

Basic Fallback Response

```
{  
  
  "intent": "Default Fallback Intent",  
  
  "responses": [  
  
    "The jungle safari begins at 6 AM. Please contact reception for bookings."  
  
  ]  
  
}
```



```
"I'm sorry, I didn't understand that. Could you rephrase?",
```

```
"Please try asking in a different way. I'm here to help!"
```

```
]
```

```
}
```

This source code reflects the structure of the basic chatbot intents used for testing during the internship.

Learning Experience

My internship experience at Ravihans & Sons Pvt. Ltd. was extremely beneficial for both my technical and personal development. Even though I was working on a guided version of the CRM chatbot, I learned how real systems are designed, structured, and tested in a professional environment.

One of the most important things I learned was how Natural Language Processing (NLP) works in practical systems. Understanding how intents, entities, and training phrases affect the accuracy of a chatbot helped me see the importance of data quality. I also learned how CRM platforms support customer interaction by storing information, tracking queries, and connecting different business workflows.

On a personal level, the internship taught me how to approach problem-solving more practically. Instead of focusing only on the technical side, I learned to think from the user's perspective — whether the response is clear, whether the system feels smooth, and how users might interact with the chatbot in real situations.

I also became more confident with documentation. Preparing reports, writing observations, and organizing my work made me understand how important documentation is in real industry settings. Overall, the experience gave me clarity about the AI field and boosted my interest in building better, smarter systems in the future.

SWOT Analysis

Strengths

- Gained a foundational understanding of how chatbots work in real business environments.
- Improved knowledge of intents, entities, NLP concepts, and testing workflows.
- Learned basic CRM integration concepts and how automation reduces operational load.
- Enhanced documentation, analysis, and observation skills through the structured internship tasks.

Weaknesses

- Limited hands-on work with full integration due to time constraints.
- Only worked on a simplified version of the chatbot and not a complete production system.
- Need to improve coding experience for backend automation and API integrations.
- Confidence in deploying live systems is still developing.

Opportunities

- Rapid growth in AI and automation offers strong future career potential.
- Opportunity to develop advanced chatbots with personalization, recommendation engines, and voice support.
- Scope to learn more CRM platforms, API integration, and backend development.
- Real-world hospitality applications provide many areas for innovation.

Threats

- Fast-changing AI technology requires continuous learning.
- Competition in AI and software roles means stronger technical expertise is needed.
- Incorrect system design or limited testing can lead to poor customer experience.
- Dependence on third-party platforms (Dialogflow, CRM tools) may limit control or flexibility.